OTEPIC

SEED COLLECTION & SAVING

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TRAINING MANUAL HANDBOOK

SEED PROPAGATION

SEED COLLECTION & SAVING

What is a seed?

It is a dormant embryo; a living, resting plant whose life processes is operating very slowly. It contains all the necessary instructions for the making of a new plant as well as a reserve supply of carbohydrates, fats, proteins, and minerals to nourish the dormant, encased seedling.

Kinds of seeds

1. Dicots-majority of flowering plants. They have two seed leaves or cotyledons which store food for the young germinating plant for example-legumes, brassicas, and Solanums.
2. MONOCOTS-contain only one cotyledon. Food is stored in the endosperm tissue which surrounds the embryo, rather than leaves.

Seed formation

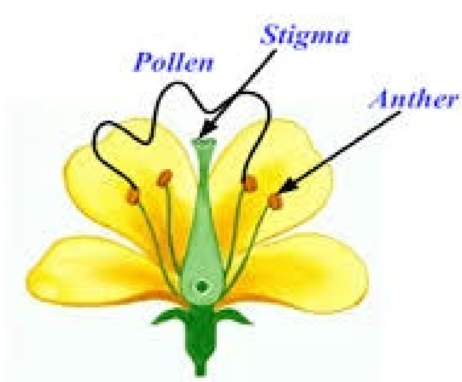
Every flower's purpose in life is to produce seed. Though differing greatly in many respects, each contains the two essential ingredients for manufacturing seed.

Stamen-the "male", pollen bearing part of a plant. Containing a long thin stalk (filament) and pollen sacs (anthers).

Pistil-the "female" organ which receives pollen and nurtures future seeds.

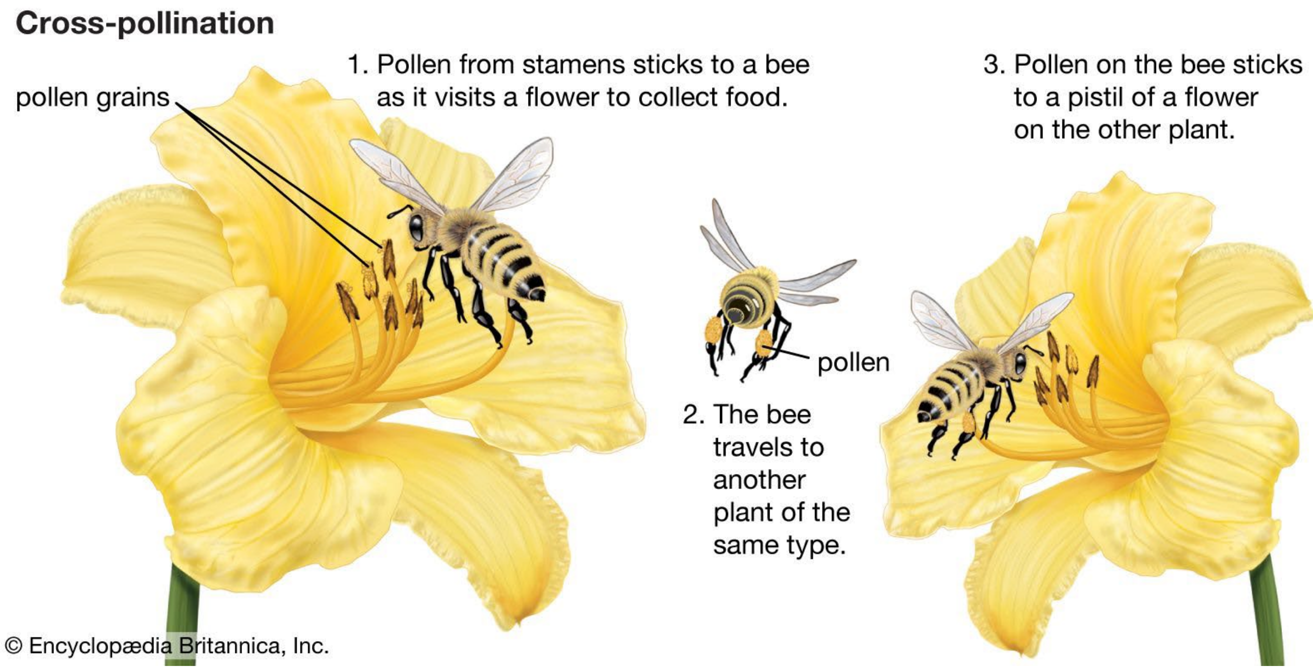
Types of pollination

1. **Self-pollination-** The flower is capable of pollinating itself with the aid of wind or insects.



Self pollination

2. **Cross pollination-** The pollen from one flower fertilizes another flower, either on the same or on a different plant. Pollen is carried by wind or insects (usually bees). Some cross-pollinated plants may be complete flowers yet be SELF-STERILE, thus requiring pollen from other plants for pollination.



Many of these vegetables easily cross with other varieties of their family. So, caution should be taken if saving seeds from them. Minimal distance is usually recommended to prevent undesired genetic combinations. If required separation is impossible, hand pollination may be applied. This involves caging and isolating plants you want to propagate. Specific techniques will be covered in a lesson later.

Understanding the pollination process of each of the vegetables you raise for seed is necessary for a successful seed saving program. For fertilization to take place, pollen must be of correct kind and it must arrive at the right time, when the plant's parts are well enough developed to enable it to reproduce. Pollen from an unrelated species will be rejected. Some plants will not receive pollen from other plants in their species. Some flowers can discharge pollen before their stigma is ready to receive it while others are simply sterile to their own pollen. Fertilization is fascinating, complex process posing an exciting challenge to anyone interested in seed production.

Why save seeds?

There are many good reasons. Here are a few: economics, security, increased self-reliance, production of plants best suited to your climate and conditions, preservation on genetic diversity, gaining knowledge, personal satisfaction and adventure.

Careful selection is the key to a successful seed saving program. It allows you not only to increase the quantity of garden plants but also to improve and refine their quality. Choose the superior plants whose seed will produce another generation with the same desirable characteristics.

Some qualities to consider:

Flavor, yield, color, size, vigor, storage life, disease and insect resistance, drought tolerance, good germination, early bearing (fruits, heads, etc.), late in bolting, quality of seed, texture, juiciness, etc.

PROPAGATION USING SEED FLATS

Why use seed flats?

1. Permits growing crops in succession
2. Protects small seedlings from bad weather and insect attack
3. Gardener ca control soil texture
4. Conserves water
5. Benefits the growth of fibrous rooted vegetables
6. Enables gardener to transplant only the healthiest seedlings

Sowing mix

Just like sowing seeds directly into a garden bed, the growing medium is very critical for the healthy germination of your crop. It does not need to be as rich in nutrients as a bed in which the crop will remain throughout its maturity. Young seedlings simply require a nutritious "breakfast", proper moisture, good drainage, air circulation, and the proper amount of heat and light. A well prepared, fertile flat/seedbed will result in strong, healthy seedlings. The sowing mix is a moist combination of sifted soil, compost/aged manure, and sand. Approximately 1/3 each by weight makes for a fertile, loose-textured mixture.

Preparing flat

Lay down a ¼" layer of leaf mulch or straw on the bottom of the flat. This provides and prevents loss of soil. A fine layer of crushed eggshells may be placed upon the straw for calcium loving. Next fill the flat with the moist soil mixture, tap down, slightly firm down edges, add a bit more oil, if necessary, then level flat with a long stick.

Sowing method

Broadcast your seed evenly over the surface of soil in the flat. The size of seed and plant variety will determine the proper spacing.

NURSERY BED

A nursery bed is well dug and fertilized portion of land, provided with good drainage and protected by shade, made structures. Roofs of grass, mats, polythene, etc. it is used for growing seeds or seedlings of vegetables, fruits, trees, etc. Contrary to a lath house, a nursery bed is a simple structure which takes less time, money and energy to prepare.

Kindly refer to Land Preparation Handbook for procedure of Seedbed preparation and Composting

Maintaining a nursery bed

1. Newly planted seeds need adequate moisture, shade and air. Water your seeds at least twice a day for quick germination.
2. Once seedlings attain two true leaves prick them out to a new spot for better, strong, and healthy growth. (Sow seeds thickly in your nursery bed).
3. At 4 true leaves, transplant your seedlings into a well prepared shamba.
4. Transplant seedlings after moistening the seed bed.

Offers numerous advantages for both individual farmers and broader agricultural systems. Here are the key benefits:

1. **Cost Savings**: Reduced Seed Costs: By saving seeds from year to year, farmers can significantly lower the cost of purchasing new seeds. This is especially beneficial for smallholder farmers with limited resources.
2. **Access to High-Quality Seeds:** Saving seeds allows farmers to select the best-performing plants from their own harvest, ensuring high-quality seeds suited to local conditions.
3. **Adaptation to Local Conditions**: Better Adaptation to Local Climate and Soil: Over time, saved seeds can become better adapted to the local environment, leading to improved resilience in the face of specific weather patterns, pests, or diseases.
4. **Selection for Desired Traits**: Farmers can select seeds from plants that demonstrate desirable traits such as drought tolerance, disease resistance, or higher yields, which may be more suited to their specific farming conditions.
5. **Preservation of Biodiversity**: Maintaining Plant Varieties: Saving seeds helps preserve heirloom and indigenous plant varieties that may not be commercially available. This contributes to agricultural biodiversity, which is essential for long-term sustainability.
6. **Protection from Monoculture Risks**: Relying on a single seed variety or commercial hybrid can expose crops to risks like disease or pests. Saving a wide range of seeds helps prevent the negative impacts of monocultures.
7. **Increased Food Security: Self-Sufficiency**: By saving seeds, farmers are less reliant on external sources for their future crop production. This increases food security, particularly in areas where access to seeds may be limited or expensive.
8. **Resilience in Crisis Situations**: In the face of natural disasters, economic instability, or supply chain disruptions, having a stock of saved seeds ensures continued crop production and food availability.

9. **Sustainability and Environmental Benefits**: Reduced Dependence on Commercial Seed Suppliers: Seed saving reduces the environmental impact associated with purchasing seeds, such as the carbon footprint from transportation and packaging.
10. **Preserving Traditional Knowledge**: Seed saving often involves traditional agricultural practices, fostering local knowledge, and promoting sustainable farming techniques.
11. **Fostering Autonomy and Independence**: Empowerment for Farmers: Seed saving gives farmers more control over their agricultural practices, allowing them to be less dependent on commercial seed companies or external inputs.
12. **Cultural Heritage**: In many cultures, seed saving is an important part of agricultural heritage. It allows communities to maintain connections to their farming traditions and history.
13. **Improved Crop Resilience**: Natural Selection: Over generations, saved seeds undergo a form of natural selection, favoring plants that are best suited to local growing conditions and challenges. This can lead to improved hardiness and yield potential in future crops.
14. **Pest and Disease Resistance**: By saving seeds from plants that show resistance to local pests or diseases, farmers can gradually build crops that are more resilient to those threats.
15. **Supporting Local Economy**: Local Seed Exchange: Saved seeds can be shared or traded with other farmers in the community, promoting local seed networks and strengthening rural economies.
16. **Market Differentiation**: Farmers who save and grow unique or heirloom varieties may find niche markets for their crops, where consumers are willing to pay a premium for rare or culturally significant varieties.
17. **Preserving Genetic Diversity**: Maintaining Genetic Resources: Seed saving helps preserve the genetic diversity of plant species, which is crucial for ensuring the availability of genetic material for future breeding programs and for coping with climate change.
18. **Prevention of Genetic Erosion**: Commercial seed systems often focus on a limited number of high-yielding varieties. Saving a diverse array of seeds helps combat genetic erosion by maintaining a broader gene pool.

In summary, seed saving offers practical and ecological benefits, from reducing costs and increasing crop resilience to preserving biodiversity and fostering food sovereignty. It plays a crucial role in sustainable agriculture and long-term food security.

TRANSPLANTING

Definition: The transplanting of seedling to a second flat, nursery area, or garden bed in order to provide adequate space and essential nutrients for optimal growth.

For a plant to undergo a healthy transplant, it roots must be able to provide adequately for top growth.

To help reduce transplant shock:

1. Don’t let the seedlings get too much manure.
2. Expose roots to the air for shortest time possible.
3. Carry soils with roots.
4. Minimize handling of seedlings and handle them gently.
5. Harden off” the seedlings by reducing watering and introducing them to slightly harsher environmental conditions several days before transplanting.
6. Place seedlings into a more nutritious, moist flat/garden bed.
7. Transplant in the early evening, or on a cool, cloudy day, and avoid windy, very dry conditions.

8. Provide shade if necessary.

Procedure for transplanting

Press down firmly, but not too tightly, allowing for aeration, water penetration and nutrient uptake. As soon as possible, give the seedlings a gentle yet thorough watering, enough to settle the soil around the roots, eliminate excess air space, and provide an adequate amount of water for growth. Again, if one can provide the seedlings with some form of sun and wind protection for days, a more successful transplant will result.

Seed Collection gallery